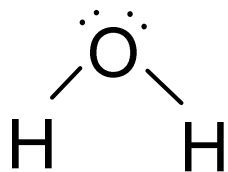


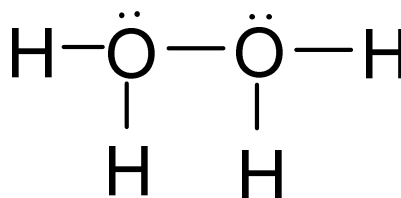
Empirical Formula

The **empirical formula** is the simplest (reduced) version of the molecular formula. It expresses the ratio of atoms in a compound in small whole numbers.

The **molecular formula** is *true* formula. It contains the actual number of atoms in a compound.

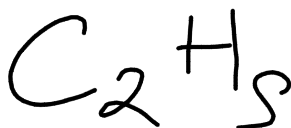


Empirical formula

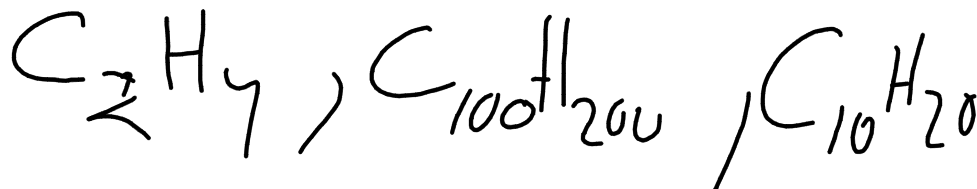


Molecular formula

If the MF is C_4H_{10} , what is the EF?



If the EF is CH_2 , what are some possible MFs?



How to determine the EF?

You will need either the percent composition or the mass of the substance

- 1) Convert % comp. into mass (assume 100 g)
- 2) Convert mass to moles
- 3) Divide by the smallest number of moles
- 4) Multiply to make whole

Ex) A compound consists of 85.7% carbon and 14.3% hydrogen.
What is the EF?

$$\textcircled{1} \text{Mass C} = .857(100\text{g}) = 85.7\text{g}$$
$$\text{H} = .143(100\text{g}) = 14.3\text{g}$$

$\textcircled{2}$ moles

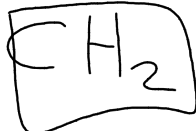
$$n = \frac{m}{M}$$

$$\text{C} = \frac{85.7\text{g}}{12.01\text{g/mol}} = 7.14\text{mol}$$

$$\text{H} = \frac{14.3\text{g}}{1.01\text{g/mol}} = 14.2\text{mol}$$

$$\text{C} = \frac{7.14\text{mol}}{7.14\text{mol}} \quad \text{H} = \frac{14.2\text{mol}}{7.14\text{mol}}$$

$$= 1.00 \quad = 2.00$$



Ex) What is the EF of a compound containing 52.1% carbon, 34.7% oxygen and 13.2% hydrogen?

$$\begin{aligned}
 C &= .521(100) & C &= \frac{52.1}{12} \\
 &= 52.1g & &= 4.34 \text{ mol} \\
 H &= .132(100) & O &= \frac{34.7}{16} \\
 &= 13.2g & &= 2.17 \text{ mol} \\
 O &= .347(100) & & \\
 &= 34.7g & H &= \frac{13.2}{1} \\
 & & & \\
 \frac{4.34 \text{ mol}}{2.17 \text{ mol}} & O = \frac{2.17 \text{ mol}}{2.17 \text{ mol}} & & \\
 C = 2 & = 1 & & \\
 H &= \frac{13.2}{2.17} & & \\
 &= 6 & & \\
 & & C_2O_2H_6 & \\
 & & EF &
 \end{aligned}$$

Ex) Phosphorus combines with oxygen to form an oxide containing 43.6% oxygen. What is the EF of this oxide?

$$\begin{aligned}
 \textcircled{1} P &= .564(100g) \\
 &= 56.4g \\
 O &= .436(100g) \\
 &= 43.6g \\
 \textcircled{2} \text{ moles} & \\
 n_P &= \frac{56.4g}{30.97 \text{ g/mol}} \\
 &= 1.82 \text{ mol} \\
 n_O &= \frac{43.6g}{16.00 \text{ g/mol}} \\
 &= 2.72 \text{ mol} \\
 \textcircled{3} P &= \frac{1.82 \text{ mol}}{1.82 \text{ mol}} & O &= \frac{2.72 \text{ mol}}{1.82 \text{ mol}} \\
 &= 1 & &= 1.5 \\
 & & & \\
 & & 2 & 3 \\
 & & P_2O_3 &
 \end{aligned}$$